

Data Preprocessing

```
In [12]: import numpy as np          #Array
import matplotlib.pyplot as plt
import pandas as pd
```

IMPORT THE DATASET

```
In [13]: dataset = pd.read_csv(r"C:\Users\sande\OneDrive\Desktop\Naresh IT\data science and
dataset
```

```
Out[13]:
```

	State	Age	Salary	Purchased
0	Mumbai	44.0	72000.0	No
1	Bangalore	27.0	48000.0	Yes
2	Hyderabad	30.0	54000.0	No
3	Bangalore	38.0	61000.0	No
4	Hyderabad	40.0	NaN	Yes
5	Mumbai	35.0	58000.0	Yes
6	Bangalore	NaN	52000.0	No
7	Mumbai	48.0	79000.0	Yes
8	Hyderabad	50.0	83000.0	No
9	Mumbai	37.0	67000.0	Yes

INDEPENDENT VARIABLE

```
In [14]: X = dataset.iloc[:, :-1].values
```

DEPENDENT VARIABLE

```
In [15]: y = dataset.iloc[:, 3].values
```

SKLEARN FILL MISSING NUMERICAL VALUE

```
In [16]: from sklearn.impute import SimpleImputer
imputer = SimpleImputer()
imputer = imputer.fit(X[:,1:3])
X[:, 1:3] = imputer.transform(X[:,1:3])
```

IMPUTE CATEGORICAL VALUE FOR INDEPDENT

```
In [17]: from sklearn.preprocessing import LabelEncoder
labelencoder_X = LabelEncoder()
labelencoder_X.fit_transform(X[:,0])
X[:,0] = labelencoder_X.fit_transform(X[:,0])
```

IMPUTE CATEGORICAL VALUE FOR DEPENDENT

```
In [18]: labelencoder_y = LabelEncoder()
y = labelencoder_y.fit_transform(y)
```

SPLIT THE DATA

```
In [19]: from sklearn.model_selection import train_test_split
X_train,X_test,y_train,y_test = train_test_split(X, y, test_size = 0.2, random_stat
```